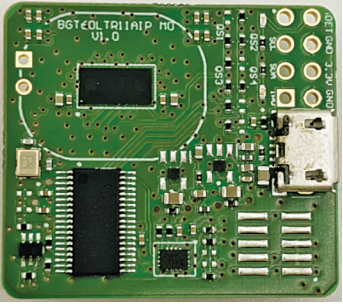
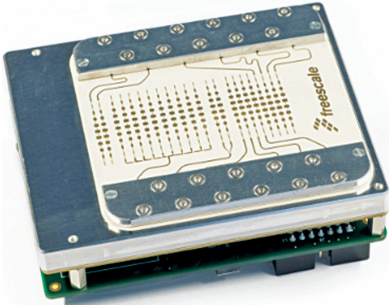


Interesting Reference Designs Of RADAR

Title of Reference Design	Radar Sensor	2D Radar Transceiver
Image		
Key Attractions	The reference design offers motion detection, direction sensing, and smart power-saving in a compact design. It works through non-metallic materials, supports custom processing, and operates efficiently, making it suitable for battery-powered and embedded applications.	The 2D radar design offers six receiving and four transmitting channels, accurate signal processing, Ethernet connectivity, customisable hardware, and software support. It operates at 76-77GHz with flexible beam control and an open-frame design.
Highlights	The reference design includes Infineon's BGT60LTR11AIP 60GHz radar sensor and an XMC1302 microcontroller with an ARM Cortex-M0, allowing users to test the sensor and create radar-based applications on a low-power chip. It is a small motion sensor with built-in antennas and a system for stable operation, making it a good alternative to PIR sensors, especially for battery-powered devices. It also has built-in detectors that give simple motion and direction outputs, and users can collect raw radar data for custom processing. Since radar sensors can work through plastic and other non-metallic materials, they can be hidden inside products. The board is small and easy to use, as it does not require an extra antenna design. It can detect movement and whether a person is approaching or moving away. It can work independently or connect to an external Cortex-M0 microcontroller for more processing. A 38.4MHz crystal provides a steady clock for the sensor, and there are connectors for debugging and accessing key signals. The board can detect human movement up to 14 metres away with an advanced sensing algorithm.	The MR2001_RD reference design from RFbeam Microwave helps engineers build 2D radar systems for cars, industry, and signal processing. It uses NXP's MR2001 chipset, which has six receiving channels and a low-noise transmitter for accurate signals. The radar creates frequency sweeps with a digital converter and processes signals to detect distance and speed, updating ten times per second. Data is sent through an Ethernet connection for easy communication. Engineers can change antennas and add their own processing hardware. The system includes MATLAB software, C-code firmware, and a user interface to select radar modes and view targets. An open-loop frequency system keeps it stable, and users can switch between wide and narrow beams for different situations. This makes the MR2001_RD a flexible and accurate radar solution.
Applications	The reference design enables motion-based wake-up and sleep in laptops, tablets, TVs, and speakers. It is suitable for battery-powered devices.	The reference design suits automotive radar, industrial sensing, and advanced signal processing applications.
OEM Brand	Infineon Technologies	RFbeam Microwave
URL	https://www.electronicsforu.com/electronics-projects/radar-sensor-reference-design	https://www.electronicsforu.com/electronics-projects/2d-radar-transceiver-reference-design